BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

MATHEMATICS | SSC-I (9th Grade) | Units 1, 2, 4, 5, 7 | Marking Report

Exam: exam-14-ch1-2-4-5-7 (Version 1) | Candidate: Seemab | Date: 29 July 2026 | Total Marks: 65

Marked and independently re-verified: every Section B and Section C line was re-derived from scratch against the exam paper (not just letter-matched to the answer key), and all 12 Section A letters were checked one by one against the key. Q.12 (quadrant of $(-x, y)$) was cross-checked twice at 400 DPI before being confirmed as a genuine miss — it ends Seemab's 3-exam perfect MCQ streak. Final verified score: **63/65 (96.9%, A+)**.

63 / 65 96.9% A+

Section A: 11/12 | Section B: 32/33 | Section C: 20/20 (perfect) | 2 marks lost across 2 questions

11/12	32/33	20/20
SECTION A: MCQS	SECTION B: SHORT	SECTION C: LONG
91.7%	97.0%	100% — perfect

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question (abbreviated)	Correct	Seemab	Result
1	Property that gives $b = a$ from $a = b$	B (Symmetric)	B	CORRECT
2	$ac < bc, c > 0 \Rightarrow a < b$, by which property	B (Cancellation, mult.)	B	CORRECT
3	Scientific notation of 0.0000728	A (7.28×10^{-5})	A	CORRECT
4	Digits in 2^{60} (log 2 = 0.3010)	C (19)	C	CORRECT
5	Antilog of 3.0000	C (1000)	C	CORRECT
6	Factors of $x^2 + 2x - 15$	A ($(x+5)(x-3)$)	A	CORRECT
7	Factorization of $8 + y^3$	A ($(2+y)(4-2y+y^2)$)	A	CORRECT
8	HCF=$(x-3)$, LCM=$(x-3)(x+2)(x-5)$: product P×Q	A ($(x-3)^2(x+2)(x-5)$)	A	CORRECT
9	Solution set of $\lvert 6 - 3y\rvert = 0$	A ($\{2\}$)	A	CORRECT
10	Solution of $-4x \geq 20$, $x \in R$	A ($x \leq -5$)	A	CORRECT
11	Quadrant of $(-4, -(-6))$	B (Second)	B	CORRECT
12	Quadrant of $(-x, y)$ when $x < 0$, $y > 0$	A (First)	B	WRONG

MCQ score: 11/12 (91.7%). One miss: Q.12 — with $x < 0$, $-x$ becomes positive, and combined with $y > 0$ the point $(-x, y)$ has both coordinates positive, placing it in the **First** quadrant (A). Seemab chose B (Second quadrant), possibly carrying over the previous question's "Second quadrant" answer without re-deriving the sign of $-x$ fresh. This ends a 3-exam perfect MCQ streak (Biology exam-12, Chemistry exam-16, Physics exam-11) and is Math's 3rd MCQ miss in 10 attempts (after exam-07 and exam-13).

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)

All 11 parts attempted. OR variant chosen for 2(i), 2(iii), 2(iv), 2(vi), 2(vii), 2(viii), 2(x); Main chosen for 2(ii), 2(v), 2(ix), 2(xi).

Q.2(i) OR: Fabric Cost per Yard 3/3

Chose: OR alternative

Cost per yard = total price ÷ yards = 139.2 ÷ 3.2 = **Rs. 43.50**. Correct setup and arithmetic, matches the key exactly.

Q.2(ii) Main: Total Balloon Cost 3/3

Chose: Main

42 balloons × Rs. 1.85 each = **Rs. 77.70**. Correct.

Q.2(iii) OR: Wajid's Average Speed & Reasonableness 3/3

Chose: OR alternative

Actual average = 283.4 ÷ 16.2 ≈ **17.49 km/litre**. Since 17.49 is very close to the claimed 17.5, concluded "It's reasonable." Matches the key precisely, including the reasonableness judgement.

Q.2(iv) OR: Antilog of 3.5636 and 2.9281 3/3

Chose: OR alternative

(a) Mantissa .5636 → table value 3661, characteristic 3 → antilog 3.5636 ≈ **3661**. (b) Mantissa .9281 → table value 8474, characteristic 2 → antilog 2.9281 ≈ **847.4**. Both correct, full working shown.

Q.2(v) Main: Kalma Pak Timing, Converted to Days 2/3

Chose: Main

Total time = 5×10^6 = 5,000,000 s → 5,000,000 ÷ 3600 ≈ **1388.8 hours** → 1388.8 ÷ 24 ≈ 57.86 days.

*−1: the question explicitly instructs to "round down" and "express in standard form." 57.86 rounds DOWN to 57 days (≈ 5.7×10^1 days), but Seemab rounded UP to **58 days** and never wrote the standard-form figure. The division itself is correct throughout — only the final rounding direction and the requested standard-form step are wrong.*

Q.2(vi) OR: Factorize $2x^2 - 5x - 3$ 3/3

Chose: OR alternative

Split middle term: $2x^2 - 6x + x - 3 = 2x(x-3) + 1(x-3) =$ **$(2x + 1)(x - 3)$**. Correct, matches the key exactly.

Q.2(vii) OR: HCF of $2x^2+5x-3$ and x^2+2x-3 3/3

Chose: OR alternative

$2x^2+5x-3 = (2x-1)(x+3)$. $x^2+2x-3 = (x+3)(x-1)$. Common factor → **HCF = $(x + 3)$**. Both factorizations correct, matches key exactly.

Q.2(viii) OR: Absolute Value Equation $3|z-2|-4=-2$ 3/3 — *Chose: OR alternative*

$3|z-2| = 2 \rightarrow |z-2| = 2/3 \rightarrow z-2 = \pm 2/3 \rightarrow$ **z = 8/3 or z = 4/3**. Solution set matches the key exactly, though the working carries a couple of unnecessary extra multiply-by-3 steps that don't affect the correct result.

Q.2(ix) Main: Compound Inequality ($3x+21<1-x$ or $3x+8>3-2x$) 3/3 — *Chose: Main*

$4x < -20 \Rightarrow x < -5$; $5x > -5 \Rightarrow x > -1$. Union: **{x | x ∈ R, x < −5 or x > −1}**. Matches the key exactly.

Note (no marks affected): Seemab's handwriting marks negative numbers with a small dash above the digit rather than to its left (the same convention she uses for negative exponents, e.g. 10^{-5}). This was legible on close inspection, but a different examiner unfamiliar with the habit could misread it — worth switching to a standard minus sign directly to the left for exam clarity.

Q.2(x) OR: Rectangle ABCD Missing Vertices 3/3 — *Chose: OR alternative*

A = (0, 0), C = (−4, 2), with B(0,2) and D(−4,0) given. Correctly plotted on a labelled graph. Matches the key exactly.

Q.2(xi) Main: Midpoint — Find C given A, B 3/3 — *Chose: Main*

B(−1,7) is the midpoint of A(3,4) and C(x,y). $(3+x)/2=-1 \Rightarrow x=-5$. $(4+y)/2=7 \Rightarrow y=10$. **C = (−5, 10)**. Correct, matches the key exactly.

Section B total: 32 / 33 (97.0%) — 10 of 11 questions fully correct; 1 mark lost on Q.2(v)'s rounding direction and missing standard form.

SECTION C — Long Answer Questions (4 × 5 = 20 Marks)

All 4 long questions attempted, every sub-part completed. Main chosen for 3(i), 3(iii); OR chosen for 3(ii), 3(iv).

Q.3(i) Main: Evaluate log(63/20) & Digits in 6^{45} 5/5 — *Chose: Main*

(a) $63/20 = (3^2\times7)/(2^2\times5) \rightarrow \log = 2\log3+\log7-2\log2-\log5 = 2(0.4771)+0.8451-2(0.3010)-0.6990 =$ **0.4983**. (b) $\log 6 = \log2+\log3 = 0.7781$; $\log(6^{45}) = 45\times0.7781 = 35.0145 \rightarrow$ characteristic 35 → **digits = 36**. Both parts fully correct with complete working, matches the key exactly.

Q.3(ii) OR: Machine Surface Area $25x^2-30x+9$ 5/5 — *Chose: OR alternative*

Area $= (5x)^2-2(5x)(3)+3^2 =$ **$(5x-3)^2$**. (a) Side = $(5x-3)$ m. (b) Boundary = $4(5x-3) = (20x-12)$ m. (c) Cost of polishing = $75\times(25x^2-30x+9) =$ Rs.$(1875x^2-2250x+675)$. (d) Cost of edging 2 sides = $28\times2(5x-3) =$ Rs.$(280x-168)$. All four parts correct, matches the key exactly.

Q.3(iii) Main: Double Inequality & Absolute Value 5/5 — *Chose: Main*

(a) $1\le5-3x\le22$ → subtract 5: $-4\le-3x\le17$ → divide by −3 (reverse): **$-17/3 \le x \le 4/3$**. Number line described correctly: solid dots at −17/3 and 4/3, shaded segment between them. (b) $|5y|=9 \rightarrow$ **y = 9/5 or y = −9/5**. Both parts fully correct, matches the key exactly.

Q.3(iv) OR: Rhombus Proof A(5,8), B(7,5), C(5,2), D(3,5) 5/5 — *Chose: OR alternative*

AB = BC = CD = DA = $\sqrt{13}$ (all four sides computed correctly) → "all sides (four) are equal so ABCD is rhombus." Diagonals: AC = $\sqrt{36}$ = **6**, BD = $\sqrt{16}$ = **4**. All eight distance calculations correct, matches the key exactly.

Section C total: 20 / 20 (100%, perfect) — all 4 long questions fully correct with complete working on every sub-part.

Complete Deduction Log (reconciles to headline score)

Question	Awarded	Lost	Reason
Q.12 (MCQ) — Quadrant of (−x, y)	0/1	−1	Chose B (Second quadrant). With x<0, −x>0; combined with y>0, both coordinates of (−x,y) are positive, so the correct quadrant is A (First).
Q.2(v) Main — Kalma Pak Timing	2/3	−1	Division fully correct (57.86 days), but the question's explicit "round down" instruction was not followed (rounded up to 58) and the requested standard-form figure (5.7×10^1 days) was never written.
Total deductions		**−2**	**65 − 2 = 63/65 confirmed. Every other Section A, B, and C question scored full marks — both deductions come from instruction-following slips (a carried-over MCQ answer, and a rounding-direction/standard-form miss), not conceptual errors.**

Strengths in This Attempt

- **Perfect Section C (20/20)** — all four long questions fully correct, including the hardest item on the paper (log(63/20) via prime-factor decomposition and digit-count of 6^{45}) and a complete 8-distance rhombus proof (4 equal sides + both diagonals).
- **Strong Section B (32/33)** — 10 of 11 questions scored full marks, including both antilog lookups, the compound inequality, and correctly identifying rectangle vertices A and C from B and D.
- **Clean OR/Main selection discipline** — every one of the 15 Section B/C questions clearly labelled which version was attempted, with no main/OR content mix-ups anywhere on the paper.
- **No incomplete sub-parts anywhere** — unlike the file-wide recurring pattern seen in other subjects, every multi-part question (machine surface (a)-(d), double inequality + absolute value, rhombus sides + diagonals) was finished in full.

Areas to Revise

- **Follow explicit rounding-direction instructions literally (−1 mark, new observation):** Q.2(v) computed 57.86 days correctly but rounded to the nearest integer (58) instead of rounding DOWN as the question specifically asked, and the requested standard-form answer was skipped. Drill fix: when a question says "round down," treat it as a floor operation, not ordinary rounding, and always finish with any explicitly-requested format (standard form, units, etc).
- **Re-derive quadrant signs fresh on each MCQ (−1 mark):** Q.12 chose the same "Second quadrant" answer as Q.11 just above it. Drill fix: re-evaluate the sign of every coordinate from scratch for each question, even when a nearby question looks similar.
- **Negative-sign notation:** writing the minus as a small dash above the digit (as in 10^{-5}) instead of to its left is consistently legible here but is non-standard; a plain minus sign to the left avoids any risk of misreading by an examiner.

Math Exam Comparison (Units 1, 2, 4, 5, 7 Series and Related Papers)

Exam	Date	Score	%	MCQ	Sec B	Sec C	Grade
exam-01-ch1-2	18 Apr 2026	65/65	100%	12/12	33/33	20/20	A+
exam-03-ch4-5-7	22 Apr 2026	60.5/65	93.1%	12/12	29.5/33	19/20	A+

exam-04-ch4	25 May 2026	57/65	87.7%	12/12	26/33	19/20	A
exam-07-ch8	5 Jun 2026	64/65	98.5%	11/12	33/33	20/20	A+
exam-10-ch1-2-4-5-7-10	26 Jun 2026	61/65	93.8%	12/12	32/33	17/20	A+
exam-11-ch1-2-4-5-7 (1st)	3 Jul 2026	58.5/65	90.0%	12/12	31.5/33	15/20	A+
exam-11-ch1-2-4-5-7 (2nd)	5 Jul 2026	63.5/65	97.7%	12/12	32/33	19.5/20	A+
exam-12-ch1-2-4-5-7	9 Jul 2026	62/65	95.4%	12/12	31.5/33	18.5/20	A+
exam-13-ch1-2-4-5-7	17 Jul 2026	58.5/65	90.0%	11/12	32/33	15.5/20	A+
exam-14-ch1-2-4-5-7 (this)	**29 Jul 2026**	**63/65**	**96.9%**	**11/12**	**32/33**	**20/20**	**A+**

exam-14 (63/65, 96.9%) is the 2nd-highest score among the four identical exam-11-through-14 papers sharing this exact Units 1,2,4,5,7 syllabus (behind only exam-11's 2nd-attempt 97.7%), and its Section C (20/20) is the best Section C result of the four. Math subject average rises to **94.3% (613.0/650, 10 attempts)**, narrowing the gap to Chemistry (94.5%) to just 0.2 points, still trailing Physics (94.8%).